

Problem 1. Alice and Bob play a game with a string of 2017 pearls. In each move, one player cuts the string between two pearls and the other player chooses one of the resulting parts of the string while the other part is discarded. In the first move, Alice cuts the string. Thereafter, the players take turns. A player loses if he or she obtains a string with a single pearl such that no more cuts are possible. Which player has a winning strategy?

Problem 2. (IMOSL 2009 C1) Consider 2009 cards, each having one gold side and one black side, lying in parallel on a long table. Initially all cards show their gold sides. Two players, standing by the same long side of the table, play a game with alternating moves. Each move consists of choosing a block of 50 consecutive cards, the leftmost of which is showing gold, and turning them all over, so those which showed gold now show black and vice versa. The last player who can make a legal move wins.

1. Does the game necessarily end?
2. Does there exist a winning strategy for the starting player?

Problem 3. (Israel National Olympiad 2014 Q4) We are given a row of $n \geq 7$ tiles. In the leftmost 3 tiles, there is a white piece each, and in the rightmost 3 tiles, there is a black piece each. The white and black players play in turns (the white starts). In each move, a player may take a piece of their color, and move it to an adjacent tile, so long as it's not occupied by a piece of the same color. If the new tile is empty, nothing happens. If the tile is occupied by a piece of the opposite color, both pieces are destroyed (both white and black). The player who destroys the last two pieces wins the game. Which player has a winning strategy, and what is it? (The answer may depend on n)

Problem 4. (Rioplátense Olympiad 2016 level 3 P1) Ana and Beto play against each other. Initially, Ana chooses a non-negative integer N and announces it to Beto. Next Beto writes a succession of 2016 numbers, 1008 of them equal to 1 and 1008 of them equal to -1 . Once this is done, Ana must split the succession into several blocks of consecutive terms (each term belonging to exactly one block), and calculate the sum of the numbers of each block. Finally, add the squares of the calculated numbers. If this sum is equal to N , Ana wins. If not, Beto wins. Determine all values of N for which Ana can ensure victory, no matter how Beto plays.

Problem 5. (XIX Olimpiáda Matemática Rioplátense 2010) Alice and Bob play the following game. To start, Alice arranges the numbers $1, 2, \dots, n$ in some order in a row and then Bob chooses one of the numbers and places a pebble on it. A player's turn consists of picking up and placing the pebble on an adjacent number under the restriction that the pebble can be placed on the number k at most k times. The two players alternate taking turns beginning with Alice. The first player who cannot make a move loses. For each positive integer n , determine who has a winning strategy.

Problem 6. (Putnam 2020 B2) Let k and n be integers with $1 \leq k < n$. Alice and Bob play a game with k pegs in a line of n holes. At the beginning of the game, the pegs occupy the k leftmost holes. A legal move consists of moving a single peg to any vacant hole that is further to the right. The players alternate moves, with Alice playing first. The game ends when the pegs are in the k rightmost holes, so whoever is next to play cannot move and therefore loses. For what values of n and k does Alice have a winning strategy?

Problem 7. (2020 Argentina OMA L3 p6) Let $n \geq 3$ be an integer. Lucas and Matías play a game in a regular n -sided polygon with a vertex marked as a trap. Initially Matías places a token at one vertex of the polygon. In each step, Lucas says a positive integer and Matías moves the token that number of vertices clockwise or counterclockwise, at his choice. The two players know the value of n and the position of the trap.

1. Determine all the $n \geq 3$ such that Matías can locate the token and move it in such a way as to never fall into the trap, regardless of the numbers Lucas says. Give the strategy to Matías.
2. Determine all the $n \geq 3$ such that Lucas can force Matías to fall into the trap. Give the strategy to Lucas.

Problem 8. (Tournament of Towns 2020 oral p6) There is an endless supply of white, blue and red cubes. In a circle arrange any N of them. The robot, standing in any place of the circle, goes clockwise and, until one cube remains, constantly repeats this operation: destroys the two closest cubes in front of him and puts a new one behind him a cube of the same color if the destroyed ones are the same, and the third color if the destroyed two are different colors. We will call the arrangement of the cubes good if the color of the cube remaining at the very end does not depend on where the robot started. We call N successful if for any choice of N cubes all their arrangements are good. Find all successful N .